

Day 10 Evidence Workbooklet

Week 2 Decision Evidence

Engineering practice → branch map → case table → test suite → Week 3 bridge

Student copy. Guided workbooklet, not the mastery quiz. Print format: duplex, left-stapled packet.

Name		Section		Date	
Score	____/35		Mastery check	secure / developing / needs support	

Day 10 closes the Week 2 sensor-kit checkout scenario. The team needs a route map, a case table, tests, and a short decision note before using the checkout process on many kits. The focus is consolidation: branch routing, first-true order, testing, debugging, and explaining why repeated cases will need loops next week.

Engineering task	Judge whether a sensor-kit checkout route is ready to use on real checkout records and what evidence should be kept.
Computational move	Synthesize branch maps, case traces, tests, and repair logic into a bounded decision-evidence packet.
Python focus	<code>if/elif/else</code> , first-true route, test suite, debug repair, closeout note, and bridge to repeated cases.
Evidence to keep	branch map, case trace, first-true reason, boundary test, regression test, bug cause, repair, and Week 3 bridge note.

Day 10 working rules

1. Use the route map before judging a case.
2. Stop at the first true rule.
3. Include tests for risks, boundaries, and clean release.
4. A wrong release is usually a rule-order or missing-condition problem.
5. Repeated cases are the reason Week 3 introduces loops, not a reason to start loops today.

1 Read the Week 2 release map

recognize

predict

Day 10 closes the sensor-kit checkout scenario. The goal is not new syntax. The goal is to use branch routing, ordered validation, expected-vs-actual evidence, and tests in one bounded decision surface.

```
1 if damage_note != "":
2     route = "repair bench"
3 elif charge_percent < 80:
4     route = "charge station"
5 elif calibration_label != "current":
6     route = "calibration bench"
7 elif borrower_initials == "":
8     route = "borrower log needed"
9 else:
10    route = "release kit"
```

Pts.	Prompt	My evidence
1	What route has highest priority?	
1	What route happens when the kit is low on charge and not damaged?	
1	What route happens when every hold condition is false?	
1	What Week 3 problem will appear if there are many kits to check?	

2 Complete a decision case table

[trace](#) [explain](#)

Trace each kit through the release map. Stop at the first true rule.

Pts.	Kit	Charge	Calibration	Other evidence	Route and reason
2	A	92	current	no damage, initials present	
2	B	92	current	cracked case, initials present	
2	C	78	expired	no damage, initials present	
2	D	88	current	no damage, blank initials	

Choose tests that make the release map trustworthy before the desk uses it on many kits.

Pts.	Prompt	My test
2	One test for the highest-priority route.	
2	One boundary test for charge exactly at 80.	
2	One test for an expired calibration label.	
2	One test that should release the kit.	

Test-suite warning

A test suite that only checks release-ready kits can pass even when every hold route is broken. Include risks, boundaries, and a clean release case.

4 Debug and repair a wrong release

debug implement

A damaged kit was released. Read the wrong version and identify the repair.

```
if charge_percent >= 80 and calibration_label == "current" and borrower_initials != "":
    route = "release kit"
elif damage_note != "":
    route = "repair bench"
elif charge_percent < 80:
    route = "charge station"
else:
    route = "hold for review"
```

Pts.	Prompt	My response
2	What route did the wrong version choose for a damaged but otherwise ready kit?	
2	What route should it choose?	
2	What is the smallest rule-order repair?	
2	Which test from Section 3 would catch this bug?	

5 Write the Week 2 decision evidence note

[transfer](#) [explain](#)

Field	My Week 2 evidence
Branch map summary	
Most important first-true rule	
Most important test to keep	
Likely bug to watch for	
Decision about using the route on real checkout records	

Pts.	Week 2 closeout note
5	Write a closeout note that names the route decision, cites branch-order evidence and one test, and explains why Week 3 will need repeated-case tracing.
2	Circle the support route you would use next: branch map / first-true trace / test suite / bug cause / repair / repeated cases. Name one next action: _____

Bridge to Week 3

Week 2 decided routes one case at a time. Week 3 asks what happens when the same route logic must be traced across repeated cases. That is where loops become useful.

You have already traced branch routes, first-true order, expected-versus-actual behavior, and regression tests. Use that evidence to complete the provided body for VIP Module 18 Activity 18.8. The function header is part of the notebook interface, not a new topic to memorize.

Step	Notebook move	Evidence to record
1	Run the setup cell once.	Setup completed / error seen
2	Find Activity 18.8 and its <code>%%writefile</code> cell.	File name: _____
3	Keep the <code>def</code> line and parameter names unchanged.	Parameters: _____
4	Complete only the indented calculation and ordered routes.	Boundary values used: _____
5	Run the file cell, then run the visible checks.	Checks passed: _____ / _____
6	If needed, revise one cause and rerun both cells.	Expected / observed / revision:

Partial-notebook boundary

Do not run the final Day 3 ZIP/package cell unless your instructor has assigned the complete required activity set. A partial launch may not create every file that the package cell expects.

Use the desk card

If you are unsure where to edit or what to rerun, use the Module 18 notebook-workflow desk card: setup once, edit the assigned file cell, run it, run the matching visible checks, revise, and record one piece of evidence.